BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 089 · Truth Tables

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. How can you recognise a NOT gate symbol?

Q2. 22 items fill boxes of 5. How many full boxes are there?
A. 4
B. 2
C. 5
D. 4.4

Q3. A bitmap is 60 × 20 pixels at 4 bits per pixel. Calculate the raw size in bytes, excluding metadata and compression.

Q4. Entry needs a valid ticket AND membership card. “AND is true if just one input is true.” Correct this. What happens with only a ticket?

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I Do · Read new knowledge

Key vocabulary

Truth table — Lists an output for every possible input case.

Input row — One combination of the Boolean inputs.

AND — Output 1 only when both inputs equal 1.

OR — Output 1 when at least one input equals 1.

NOT — Reverses a single Boolean input.

Output column — The results for all specified inputs.

New learning

Four combinations for two inputs: For A and B, list input pairs 00, 01, 10, 11 exactly once. Each row corresponds to one possible combination.

AND truth table: AND returns 0,0,0,1 for rows 00,01,10,11; only the final row has both inputs high.

OR truth table: OR returns 0,1,1,1; the first row is the only one without a high input.

NOT truth table: NOT has one input, so only two rows are required: 0→1 and 1→0.

Worked example

Complete the AND output column for inputs 00,01,10,11, explaining the difference in row 11. [4 marks]

1. Write rows 00,01,10,11 so no input case is missing.

2. 00 produces 0 [1] and 01 produces 0 [1].

3. 10 produces 0 [1], while 11 produces 1 [1].

4. Only 11 contains both inputs high, so it satisfies AND.

We Do · Practise with guidance

Guided 1

Complete OR outputs for input pairs 00,01,10,11 in order. [4 marks]

Worked method: 00→0 [1]; 01→1 [1]; 10→1 [1]; 11→1 [1].

Guided 2

Give NOT outputs for A=0 and A=1. [2 marks]

Worked method: NOT 0=1 [1] and NOT 1=0 [1].

Guided 3

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

For A=1,B=0, explain AND and OR outputs. [4 marks]

Worked method: AND gives 0 [1] because both are not 1 [1]. OR gives 1 [1] because one input is 1 [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Complete the condition that prints 1 only when A and B are both 1.

```
for A = 0 to 1
    for B = 0 to 1
        if ... then
            print(1)
        else
            print(0)
        endif
    next B
next A
```

write

Question 1

State the number of rows in a two-input truth table, excluding headings. [1 mark] Source: Original OCR J277-style practice question.

Question 2

Complete the output column for an OR gate on input pairs 00,01,10,11. [4 marks] Source: Original OCR J277-style practice question.

Question 3

Complete four-row AND and OR output columns and explain why the outputs differ for input 0,1. [6 marks] Source: Original OCR J277-style practice question.

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Exit Ticket · Answer before checking

A two-input truth table has four combinations.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. It is a one-input inverter, usually a triangle with a small output bubble.

Q2. A — 4

Q3. 60 × 20 × 4 = 4,800 bits; divide by 8 = 600 bytes.

Q4. AND is true only when both inputs are true. One valid ticket or card alone is not enough. OR needs at least one true input; NOT reverses a value.

I Do / We Do pages · Guided practice

Q1. 00→0 [1]; 01→1 [1]; 10→1 [1]; 11→1 [1].

Q2. NOT 0=1 [1] and NOT 1=0 [1].

Q3. AND gives 0 [1] because both are not 1 [1]. OR gives 1 [1] because one input is 1 [1].

You Do pages · Code First

```
for A = 0 to 1
    for B = 0 to 1
        if A == 1 AND B == 1 then
            print(1)
        else
            print(0)
        endif
    next B
next A
```

Expected result: 0 0 0 1

You Do pages · Independent questions

Q1. Four rows [1].

Q2. 0 [1]; 1 [1]; 1 [1]; 1 [1].

Q3. AND 0,0,0,1 [2]; OR 0,1,1,1 [2]. For 0,1, AND is 0 [1] because both are not high, while OR is 1 [1] because one input is high.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: